SSC MTS 2024 question paper with answers

8 October 2024 · Shift 1

75 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 15 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 12 questions

Q.3

A thief saw a policeman and he started running at a speed of 12 m/s. After a certain time, the policeman starts running behind the thief at a speed of 18 m/s. If the policeman covered a 540 m distance to catch the thief, the time (in seconds) after which the policeman started running is:

(A) 12
(B) 10
(C) 15
(D) 18

Q.7

A flat worth ₹9,50,000 is sold by Lakshman to Bharath at 8% profit. If Bharath sells it back to Lakshman at 3% loss, then find the gain/loss (in ₹) of Lakshman in the entire transaction?

(A) Lakshman gains ₹30,780
(B) Lakshman gains ₹50,200
(C) Lakshman loses ₹45,220
(D) Lakshman gains ₹45,220

Q.9

Find the HCF of 70 and 90.

(A) 5
(B) 7
(C) 2
(D) 10

Q.10

The average of the first six multiples of 7 is:

(A) 25.1
(B) 23.2
(C) 24.5
(D) 22.4

Q.11

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 3% per year rate of interest for 4 years.

(A) 400
(B) 320
(C) 360
(D) 300

Q.12

A man riding his bicycle covers 75 metres in 12.5 seconds. What is his speed (in km/hr)?

(A) 15.5
(B) 11.1
(C) 21.6
(D) 6

Q.13

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 70 kg and 39 kg is:

(A) 52.8
(B) 49.8
(C) 51.8
(D) 50.8

Q.14

The area of a rectangular field is 396 m^2, whose sides are in the ratio of 11 : 9. What is The perimeter of the rectangular field?

(A) 80 m
(B) 74 m
(C) 78 m
(D) 76 m

Q.15

There is a 21 litres mixture of milk and water. The ratio of milk and water in the mixture is 4 : 3. Find the amount of water (in litres) that should be added to this mixture so that the resulting mixture has 50% of water in it.

(A) 3
(B) 2
(C) 1
(D) 4

Q.17

The price of a computer increases by 10% in the first year, decreases by 5% in the second year, and increases by 8% in the third year. If the original price was ₹50,000, then the final price (in₹) is:

(A) 54,120
(B) 55,830
(C) 56,430
(D) 55,100

Q.19

A hemispherical bowl of internal radius 15 cm contains a liquid. The liquid is to be filled into cylindrical shaped bottles of diameter 5 cm and height 6 cm. The number of bottles required to empty the bowl is:

(A) 60
(B) 50
(C) 40
(D) 30

Q.20

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 3% per year rate of interest for 6 years.

(A) 390
(B) 490
(C) 410
(D) 450

Reasoning Ability and Problem Solving 15 questions

Q.1

What should come in place of the question mark (?) in the given series? 7 12 16 30 34 66 70 ?

(A) 142
(B) 140
(C) 138
(D) 136

Q.4

In a certain code language, ‘WORD’ is coded as ‘7135’ and ‘RISE’ is coded as ‘4658’. What is the code for ‘R’ in the given code language?

(A) 5
(B) 6
(C) 4
(D) 3

Q.5

What should come in place of the question mark (?) in the given series based on the English alphabetical order? BXL, DVN, FTP, ?, JPT

(A) GRS
(B) HRR
(C) GRR
(D) HST

Q.6

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order). Only 3 people are sitting between E and A when counted from the right of E. B is sitting 2nd to the left of A. G is sitting to the immediate left of F. Only 2 people are sitting between B and G when counted from the right of B. C is the immediate neighbour of E. What is the position of D with respect to E?

(A) 3rd from the right
(B) 2nd from the right
(C) 3rd from the left
(D) 2nd from the left

Q.7

EF26 is related to JK33 in a certain way. In the same way, TU27 is related to YZ34. To which of the following is JK28 related, following the same logic?

(A) OP35
(B) PO35
(C) PO36
(D) OP36

Q.8

Which two numbers should be interchanged to make the following equation correct? $(28 \times 2) - 16 \times 6 + 8 \times 3 + 5 \times 7 = 39$ (Note: Interchange should be done of the entire number and not individual digits of a given number.)

(A) 2 and 3
(B) 6 and 7
(C) 8 and 7
(D) 5 and 3

Q.9

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statement(s) is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: All pumpkin is mutton. All mutton are fishes. Conclusions: (I) Some fishes are mutton. (II) All pumpkins are fishes.

(A) Neither conclusion (I) nor (II) follows.
(B) Both conclusions (I) and (II) follow.
(C) Only conclusion (I) follows.
(D) Only conclusion (II) follows.

Q.11

3 is related to 15 following a certain logic. Following the same logic, 12 is related to 60. To which of the following is 5 related to, following the same logic? (Note: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 - Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 25
(B) 32
(C) 30
(D) 35

Q.12

In a certain code language, ‘BENT’ is coded as ‘4628’ and ‘EARS’ is coded as ‘5471’.What is the code for ‘E’ in the given code language?

(A) 6
(B) 4
(C) 7
(D) 5

Q.13

In a certain code language, ‘I play football’ is coded as ‘ku ch ta’ and ‘he play cricket’ is coded as ‘al bi ta’. How is ‘play’ coded in the given language?

(A) bi
(B) ta
(C) ku
(D) al

Q.14

Seven people S, E, C, T, I, O and N are sitting around a circular table, facing the centre(but not necessarily in the same order). E is sitting second to the left of N, while N is sitting second to the left of S. Only one person is sitting between S and O when counted from the right of S. C is sitting to the immediate right of E, whereas T is sitting to the immediate left of O. Who is sitting to the immediate right of I?

(A) T
(B) S
(C) C
(D) O

Q.15

What will come in the place of '?' in the following equation, if '×' and ' ÷ ' are interchanged and '–' and '+' are interchanged? 23 ÷ 18 + 150 × 50 – 44 = ?

(A) 485
(B) 515
(C) 455
(D) 545

Q.16

14 is related to 68 following a certain logic. Following the same logic, 17 is related to 83. To which of the following is 21 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 107
(B) 101
(C) 105
(D) 103

Q.17

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some plants are rocks. No vegetable is a plant. Conclusions: I) No vegetable is a rock (II) Some rocks are plants.

(A) Both conclusions (I) and (II) follow
(B) Only conclusion (II) follows
(C) Only conclusion (I) follows
(D) Neither conclusion (I) nor (II) follows

Q.19

JT79 is related to DN71 in a certain way. In the same way, KU22 is related to EO14. To which of the following is LV71 related, following the same logic?

(A) PF63
(B) PF66
(C) FP66
(D) FP63

General Awareness 25 questions

Q.1

The liquid water present in oceans, lakes and rivers gets converted to the gaseous form in the presence of sunlight in the water cycle. This step is called:

(A) Precipitation
(B) Evaporation
(C) Condensation
(D) Melting

Q.2

Which of the following statements about nutrients are correct? Statement A: C, H, O, N, P and K are macro nutrients. Statement B: Fe, Mn, Ni, Co and Mo are micronutrients. Statement C: Micronutrients are not essential for our body.

(A) Only statements B and C
(B) Only statements A and C
(C) All statements A, B and C
(D) Only statements A and B

Q.3
What is the usual treatment on repo rates by the Reserve Bank of India in case of a hawkish monetary policy stance?

(A) Increased
(B) Reduced gradually
(C) Kept the same
(D) Reduced sharply by more than 100 bps

Q.4
Who is revered as ‘Karnataka Sangeeta Pitamaha’?

(A) Gopala Dasa
(B) Jagannatha Dasa
(C) Purandara Dasa
(D) Vijaya Dasa

Q.5
In cycling, what is a 'peloton'?

(A) A cycling technique for climbing hills
(B) The finish line of a race
(C) The starting line of a race
(D) A group of riders in a tight pack during a race

Q.6
As per Census of India 2011, which of the following states has the highest percentage of Hindu population in India?

(A) Uttar Pradesh
(B) Madhya Pradesh
(C) Himachal Pradesh
(D) Bihar

Q.7
Which of the following articles of the Constitution of India mentions that every Minister and the Attorney-General of India shall have the right to speak in and participate in the proceedings of either House of Parliament?

(A) Article 88
(B) Article 87
(C) Article 86
(D) Article 89

Q.8
Below are two chemical equations: Zn (s) + $CuSO_4$ (aq) → $ZnSO_4$ (aq) + Cu (s); Pb (s) + $CuCl_2$ (aq) → $PbCl_2$ (aq) + Cu (s) Why does copper get separated from its compound form in both the cases?

(A) Copper compounds break down under high pressure.
(B) Copper is more reactive than Zinc and Lead.
(C) Zinc and Lead are unstable in element form.
(D) Zinc and Lead are more reactive than Copper.

Q.9
Who among the following is/was a famous proponent of Bharatanatyam?

(A) Sunanda Nair
(B) Kalavati Devi
(C) Rukmini Devi Arundale
(D) Kumkum Mohanty

Q.10
As of July 2023, 137 unique products under 'One District, One Product' scheme has been notified from how many districts?

(A) 538
(B) 256
(C) 713
(D) 657

Q.11
What is the height of the Kanchenjunga peak of Himalaya?

(A) 8434 metres
(B) 8798 metres
(C) 8598 metres
(D) 8175 metres

Q.12
Who among the following was India's first cricket team captain?

(A) Sunil Gavaskar
(B) K Srikant
(C) Navab Pataudi
(D) CK Nayudu

Q.13
Swachh Bharat Mission was launched on the birth anniversary of which of the following freedom fighters?

(A) Subhash Chandra Bose
(B) Chandrashekhar Azad
(C) Bhagat Singh
(D) Mahatma Gandhi

Q.14
Who were the first foreign rulers of northwest India in the post-Mauryan period?

(A) Kushanas
(B) Bactrian Greeks
(C) Sakas
(D) Parthians

Q.15
Bharatanatyam Guru _____ received the Nrithya Seva Ratna award from Indian Fine Arts Akademi, San Diego in 2023.

(A) Ratikant Mohaptra
(B) Vidhya Subramanian
(C) Sailaja
(D) Hema Rajagopalan

Q.16
As of 2023, what is the number of major public seaports in India?

(A) 16
(B) 10
(C) 12
(D) 13

Q.17
Which of the following cell organelles possesses their own DNA and can synthesise protein for their functions?

(A) Golgi complex
(B) Endoplasmic reticulum
(C) Centrosome
(D) Chloroplast

Q.18
Which author wrote the Hindi novel 'Tamas'?

(A) Bhisham Sahani
(B) Yashpal
(C) Shrilal Shukla
(D) Premchand

Q.19
'Promotion of educational and economic interests of Scheduled Castes, Scheduled Tribes and other weaker sections' comes under which Article of Indian constitution?

(A) Article 56
(B) Article 36
(C) Article 46
(D) Article 66

Q.20
The training academy for Indian Police Service officers has been named after:

(A) Jawaharlal Nehru
(B) Lal Bahadur Shastri
(C) Sardar Vallabhbhai Patel
(D) Mahatma Gandhi

Q.21
Which social reformer from South India founded the Self Respect Movement?

(A) C Iyodhee Pandithar
(B) Kandukur Veeresalingam
(C) Chembeti Sridharalu Naidu
(D) EV Ramaswamy Naicker

Q.22
In the Great Stupa at Sanchi there are _____ gateways known as ‘Toranas’ at the cardinal points to the compass.

(A) 4
(B) 2
(C) 8
(D) 6

Q.23
In which of the following years was the Census Act enacted and extended to the whole of India?

(A) 1949
(B) 1951
(C) 1948
(D) 1950

Q.24
The campaign 'Ek Tareekh, Ek Ghanta, Ek Sath' is associated with which occasion?

(A) Vivekananda Jayanti
(B) Sardar Patel Jayanti
(C) Gandhi Jayanti
(D) Ambedkar Jayanti

Q.25
Who among the following has written ‘Gita Govinda’, a Sanskrit epic?

(A) Kalhan
(B) Udayaraja
(C) Jayadeva
(D) Ramachandra

English Language and Comprehension 23 questions

Q.1
Select the most appropriate ANTONYM of the given word. Bulging

(A) Swelling
(B) Hanging
(C) Weeping
(D) Contracting

Q.2
Select the most appropriate option to fill in the blank. Never _____ ever seen a more disgusting sight.

(A) have I
(B) has I
(C) did I
(D) I have

Q.3
Select the most appropriate option to fill in the blank. Roger applied the _____ in advance as he anticipated an accident.

(A) break
(B) broke
(C) brig
(D) brake

Q.4
Select the most appropriate synonym of the given word. Blend

(A) Disperse
(B) Boil
(C) Divide
(D) Mix

Q.5

The sentence below contains errors. Select the option with the correct use of tenses. Your father will waiting for you when you reach home.

(A) Your father would waiting for you when you reach home.

(B) Your father waiting for you when you reach home.

(C) Your father will be waiting for you when you reach home.

(D) Your father should waiting for you when you reach home.

Q.6

Select the most appropriate option to fill in the blank. She was barking up the wrong _____ when she filed a complaint against ragging with the students' union instead of the university administration.

(A) disc

(B) dog

(C) pillar

(D) tree

Q.8

In the following sentence the underlined part contains as error. Alternatives to the underlined part are given as options. Select the correct alternative. The sky appeared to be “greyest” after the storm.

(A) more grey

(B) grey

(C) most grey

(D) greyer

Q.9

Select the most appropriate option to fill in the blank. Living within our environmental limits is one of the central _____ of sustainable development.

(A) principles

(B) illuminates

(C) agencies

(D) demonstrates

Q.10

Select the INCORRECTLY spelt word.

(A) Gracious

(B) Spectaculor

(C) Enthusiastic

(D) Reliable

Q.11

Select the most appropriate idiom/phrase to fill in the blank. My car _____ on the way to college yesterday.

(A) broke out

(B) broke in

(C) broke down

(D) broke up

Q.12

Select the most appropriate option that can substitute the underlined segment in the given sentence. Good nutrition is the “bedrock at child survival”, health and development.

(A) bedrock through child survival

(B) bedrock off child survival

(C) bedrock from child survival

(D) bedrock of child survival

Q.13

Select the appropriate option that can substitute the underlined word in the following sentence to make it grammatically correct. This almirah is made up of “heavier” metal.

(A) heaviest

(B) most heaviest

(C) heavy

(D) most heavier

Q.14

The following sentence has been split into four segments. Identify the segment that contains anerror in phrasal verb. Tara was late for the meeting / but somehow managed / to catch out with / everything she had missed.

(A) everything she had missed
(B) Tara was late for the meeting
(C) to catch out with
(D) but somehow managed

Q.15

Select the most appropriate word from the given options that can substitute the word ‘courage’ and make a meaningful sentence. I am not courage enough to take this risk.

(A) discouraging
(B) discourage
(C) courageous
(D) encourage

Q.16

Select the option that will improve the underlined part of the given sentence. In case no improvement is needed, select 'No improvement required'. “Monika said to her mother that, "I have passed” the examination."

(A) Monika said to her mother, "I have passed
(B) Monika said to her mother that, "I am passed
(C) No improvement required
(D) Monika said to her mother that, "she has passed

Q.18

In the following sentence the underlined part contains as error. Alternatives to the underlined part are given as options. Select the correct alternative. Playing in the field, the young “gayer” students were laughing and giggling.

(A) most gay
(B) gay
(C) gayest
(D) more gay

Q.19

Select the most appropriate ANTONYM of the underlined word in the given sentence. I don't prefer to remain alone at home in the evenings because loneliness evokes a sense of “melancholy” and grief within me.

(A) Cheerfulness
(B) Readiness
(C) Shyness
(D) Absoluteness

Q.20

Select the INCORRECTLY spelt word in the given sentence. Uluru is the world's largest monolith and Australia's recognisable natural icon with an elliptical shape and a circumfarence of around 94 kilometers.

(A) Monolith
(B) Kilometers
(C) Elliptical
(D) Circumfarence

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Traditional medicines (TMs) make (1)_____ natural products and are of great importance. Such forms of medicine as traditional Chinese medicine (TCM), Ayurveda, Kampo, traditional Korean medicine (TKM), and Unani (2)_____ natural products and have been (3)_____ all over the world for hundreds or even thousands of years, and they (4)_____ blossomed into orderly-regulated systems of medicine. In their various forms, they may have certain defects, but they (5)_____ still a valuable repository of human knowledge.

Select the most appropriate option to fill in blank number 1.

(A) used of
(B) uses of
(C) used off
(D) use of

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Traditional medicines (TMs) make (1)_____ natural products and are of great importance. Such forms of medicine as traditional Chinese medicine (TCM), Ayurveda, Kampo, traditional Korean medicine (TKM), and Unani (2)_____ natural products and have been (3)_____ all over the world for hundreds or even thousands of years, and they (4)_____ blossomed into orderly-regulated systems of medicine. In their various forms, they may have certain defects, but they (5)_____ still a valuable repository of human knowledge.

Select the most appropriate option to fill in blank number 2.

(A) employs
(B) employing
(C) employed
(D) employ

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Traditional medicines (TMs) make (1)_____ natural products and are of great importance. Such forms of medicine as traditional Chinese medicine (TCM), Ayurveda, Kampo, traditional Korean medicine (TKM), and Unani (2)_____ natural products and have been (3)_____ all over the world for hundreds or even thousands of years, and they (4)_____ blossomed into orderly-regulated systems of medicine. In their various forms, they may have certain defects, but they (5)_____ still a valuable repository of human knowledge.

Select the most appropriate option to fill in blank number 3.

(A) practises
(B) practised
(C) practice
(D) practices

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Traditional medicines (TMs) make (1)_____ natural products and are of great importance. Such forms of medicine as traditional Chinese medicine (TCM), Ayurveda, Kampo, traditional Korean medicine (TKM), and Unani (2)_____ natural products and have been (3)_____ all over the world for hundreds or even thousands of years, and they (4)_____ blossomed into orderly-regulated systems of medicine. In their various forms, they may have certain defects, but they (5)_____ still a valuable repository of human knowledge.

Select the most appropriate option to fill in blank number 4.

(A) had
(B) have
(C) has
(D) have being

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Traditional medicines (TMs) make (1)_____ natural products and are of great importance. Such forms of medicine as traditional Chinese medicine (TCM), Ayurveda, Kampo, traditional Korean medicine (TKM), and Unani (2)_____ natural products and have been (3)_____ all over the world for hundreds or even thousands of years, and they (4)_____ blossomed into orderly-regulated systems of medicine. In their various forms, they may have certain defects, but they (5)_____ still a valuable repository of human knowledge.

Select the most appropriate option to fill in blank number 5.

(A) am
(B) was
(C) is
(D) are

Answer key

Numerical and Mathematical Ability

Q.3 **(C)**	Q.7 **(A)**	Q.9 **(D)**	Q.10 **(C)**	Q.11 **(C)**	Q.12 **(C)**
Q.13 **(D)**	Q.14 **(A)**	Q.15 **(A)**	Q.17 **(C)**	Q.19 **(A)**	Q.20 **(D)**

Reasoning Ability and Problem Solving

Q.1 **(C)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(A)**	Q.7 **(A)**	Q.8 **(A)**
Q.9 **(B)**	Q.11 **(A)**	Q.12 **(B)**	Q.13 **(B)**	Q.14 **(B)**	Q.15 **(C)**
Q.16 **(D)**	Q.17 **(B)**	Q.19 **(D)**			

General Awareness

Q.1 **(B)**	Q.2 **(D)**	Q.3 **(A)**	Q.4 **(C)**	Q.5 **(D)**	Q.6 **(C)**
Q.7 **(A)**	Q.8 **(D)**	Q.9 **(C)**	Q.10 **(C)**	Q.11 **(C)**	Q.12 **(D)**
Q.13 **(D)**	Q.14 **(B)**	Q.15 **(D)**	Q.16 **(C)**	Q.17 **(D)**	Q.18 **(A)**
Q.19 **(C)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(A)**	Q.23 **(C)**	Q.24 **(C)**
Q.25 **(C)**					

English Language and Comprehension

Q.1 **(D)**	Q.2 **(A)**	Q.3 **(D)**	Q.4 **(D)**	Q.5 **(C)**	Q.6 **(D)**
Q.8 **(B)**	Q.9 **(A)**	Q.10 **(B)**	Q.11 **(C)**	Q.12 **(D)**	Q.13 **(C)**
Q.14 **(C)**	Q.15 **(C)**	Q.16 **(A)**	Q.18 **(B)**	Q.19 **(A)**	Q.20 **(D)**
Q.21 **(D)**	Q.22 **(D)**	Q.23 **(B)**	Q.24 **(B)**	Q.25 **(D)**	

Explanations

Numerical and Mathematical Ability

Q.3 · Answer (C) 15

The policeman ran 540 m in 540 ÷ 18 = 30 seconds, in which the thief ran 12 × 30 = 360 m. The thief's start of 540 − 360 = 180 m took him 180 ÷ 12 = 15 seconds.

Q.7 · Answer (A) Lakshman gains ₹30,780

Lakshman sells for 9,50,000 × 1.08 = ₹10,26,000 and buys back for 10,26,000 × 0.97 = ₹9,95,220, so he gains 10,26,000 − 9,95,220 = ₹30,780.

Q.9 · Answer (D) 10

70 = 2 × 5 × 7 and 90 = 2 × 3^2 × 5; the common factors give 2 × 5 = 10.

Q.10 · Answer (C) 24.5

The multiples 7, 14, …, 42 are evenly spaced, so their average is (7 + 42)/2 = 24.5.

Q.11 · Answer (C) 360

SI = 3,000 × 3 × 4/100 = ₹360.

Q.12 · Answer (C) 21.6

75 ÷ 12.5 = 6 m/s, and 6 × 18/5 = 21.6 km/h.

Q.13 · Answer (D) 50.8

(40 + 49 + 56 + 70 + 39)/5 = 254/5 = 50.8 kg.

Q.14 · Answer (A) 80 m

With sides 11x and 9x, $99x^2 = 396$, so x = 2 and the sides are 22 m and 18 m. Perimeter = 2 × (22 + 18) = 80 m.

Q.15 · Answer (A) 3

The mixture holds 12 litres of milk and 9 of water. For 50% water the water must equal the milk, 12 litres, so 3 litres must be added.

Q.17 · Answer (C) 56,430

50,000 × 1.10 × 0.95 × 1.08 = ₹56,430.

Q.19 · Answer (A) 60

Bowl: (2/3)π × 15^3 = 2,250π cm^3. Bottle: π × 2.5^2 × 6 = 37.5π cm^3. 2,250π ÷ 37.5π = 60 bottles.

Q.20 · Answer (D) 450

SI = 2,500 × 3 × 6/100 = ₹450.

Reasoning Ability and Problem Solving

Q.1 · Answer (C) 138

The steps alternate ×2 − 2 and +4: 7 → 12 → 16 → 30 → 34 → 66 → 70. The next is 70 × 2 − 2 = 138.

Q.4 · Answer (A) 5

R is the only letter in both words and 5 the only digit in both codes, so R = 5.

Q.5 · Answer (B) HRR

First letters move +2 (B, D, F, H, J), second letters −2 (X, V, T, R, P) and third letters +2 (L, N, P, R, T): HRR.

Q.6 · Answer (A) 3rd from the right

Facing the centre, a person's right is anticlockwise. Going right from E the order is E, C, B, D, A, G, F: A is 4th to the right of E, B is 2nd to the left of A, G is 3rd to the right of B with F on G's right, and C is E's other neighbour. D is 3rd to the right of E.

Q.7 · Answer (A) OP35

Both letters move 5 places forward and the number rises by 7: E, F → J, K and 26 → 33. So J, K → O, P and 28 → 35: OP35.

Q.8 · Answer (A) 2 and 3
Interchanging 2 and 3: (28 × 3) – 16 × 6 + 8 × 2 + 5 × 7 = 84 – 96 + 16 + 35 = 39.

Q.9 · Answer (B) Both conclusions (I) and (II) follow.
All pumpkins are mutton and all mutton are fishes, so all pumpkins are fishes (II); and as all mutton are fishes, some fishes are mutton (I). Both follow.

Q.11 · Answer (A) 25
Each number is multiplied by 5: 3 × 5 = 15 and 12 × 5 = 60. So 5 × 5 = 25.

Q.12 · Answer (B) 4
E is the only letter in both words and 4 the only digit in both codes, so E = 4.

Q.13 · Answer (B) ta
'play' is the only word in both sentences and 'ta' the only code in both, so play = ta.

Q.14 · Answer (B) S
Facing the centre, a person's right is anticlockwise. Going right from S the order is S, T, O, E, C, N, I: N is 2nd to the left of S and E 2nd to the left of N, O is 2nd to the right of S with T on its left, C is on E's right and I takes the last seat. To I's right sits S.

Q.15 · Answer (C) 455
After interchanging the signs: 23 × 18 – 150 ÷ 50 + 44 = 414 – 3 + 44 = 455.

Q.16 · Answer (D) 103
Each number is multiplied by 5 and 2 subtracted: 14 × 5 – 2 = 68 and 17 × 5 – 2 = 83. So 21 × 5 – 2 = 103.

Q.17 · Answer (B) Only conclusion (II) follows
Some plants are rocks, so some rocks are plants (II). No vegetable is a plant, but vegetables may still be rocks that are not plants, so I does not follow.

Q.19 · Answer (D) FP63
Both letters move 6 places back and the number falls by 8: J, T → D, N and 79 → 71. So L, V → F, P and 71 → 63: FP63.

General Awareness

Q.1 · Answer (B) Evaporation
Evaporation turns liquid water into water vapour with the sun's heat; condensation and precipitation follow later in the cycle.

Q.2 · Answer (D) Only statements A and B
Micronutrients are needed only in small amounts but are essential all the same, so statement C is false. A names elements needed in large amounts and B ones needed in traces.

Q.3 · Answer (A) Increased
A hawkish stance puts controlling inflation first, so the RBI tends to raise the repo rate, making borrowing dearer.

Q.4 · Answer (C) Purandara Dasa
Purandara Dasa, the 16th-century composer who laid down the graded lessons still used to teach Carnatic music, is revered as its 'pitamaha'.

Q.5 · Answer (D) A group of riders in a tight pack during a race
The peloton is the main bunch of riders in a road race, who ride close together to save energy in each other's slipstream.

Q.6 · Answer (C) Himachal Pradesh
Census 2011 counted Hindus as about 95% of Himachal Pradesh's population, the highest share of any state.

Q.7 · Answer (A) Article 88
Article 88 gives every Minister and the Attorney-General the right to speak in, and take part in the proceedings of, either House, but not to vote.

Q.8 · Answer (D) Zinc and Lead are more reactive than Copper.
These are displacement reactions: a more reactive metal displaces a less reactive one from its salt, and zinc and lead stand above copper in the reactivity series.

Q.9 · Answer (C) Rukmini Devi Arundale
Rukmini Devi Arundale did much to revive Bharatanatyam and founded Kalakshetra in Madras in 1936.

Q.10 · Answer (C) 713
Under the PM Formalisation of Micro food processing Enterprises (PMFME) scheme, the Ministry of Food Processing Industries approved One District One Product for 713 districts, covering 137 unique products.

Q.11 · Answer (C) 8598 metres
SSC's key gives 8,598 m, the height in NCERT's Class 9 table of Himalayan peaks; the figure most widely cited today is 8,586 m. None of the other options is close to either.

Q.12 · Answer (D) CK Nayudu
C K Nayudu captained India in its first Test match, against England at Lord's in 1932.

Q.13 · Answer (D) Mahatma Gandhi
The Swachh Bharat Mission was launched on 2 October 2014, Mahatma Gandhi's birth anniversary.

Q.14 · Answer (B) Bactrian Greeks
After the Mauryas, the Greeks who ruled Bactria were the first to cross the Hindu Kush and rule north-west India, before the Sakas, Parthians and Kushanas.

Q.15 · Answer (D) Hema Rajagopalan
Hema Rajagopalan, the Chicago-based Bharatanatyam guru who founded Natya Dance Theatre, received the Nrithya Seva Ratna from the Indian Fine Arts Academy, San Diego, on 1 April 2023.

Q.16 · Answer (C) 12
In 2023 India had 12 major ports, under the Union Ministry of Ports, Shipping and Waterways; the others are non-major ports under the state governments.

Q.17 · Answer (D) Chloroplast
Chloroplasts, like mitochondria, have their own DNA and ribosomes and make some of their own proteins.

Q.18 · Answer (A) Bhisham Sahani
'Tamas', set during the riots of Partition, is by Bhisham Sahni; it won the Sahitya Akademi Award in 1975.

Q.19 · Answer (C) Article 46
Article 46, a Directive Principle, asks the State to promote the educational and economic interests of the weaker sections, especially the Scheduled Castes and Scheduled Tribes.

Q.20 · Answer (C) Sardar Vallabhbhai Patel
IPS officers are trained at the Sardar Vallabhbhai Patel National Police Academy in Hyderabad.

Q.21 · Answer (D) EV Ramaswamy Naicker
E V Ramasamy, 'Periyar', started the Self-Respect Movement in Tamil Nadu in 1925.

Q.22 · Answer (A) 4
The Great Stupa has four toranas, one at each cardinal point: north, south, east and west.

Q.23 · Answer (C) 1948
The Census Act, 1948 provides for taking the census and extends to the whole of India.

Q.24 · Answer (C) Gandhi Jayanti
'Ek Tareekh, Ek Ghanta, Ek Saath' was an hour of nationwide cleaning on 1 October 2023, a tribute to Mahatma Gandhi on the eve of Gandhi Jayanti.

Q.25 · Answer (C) Jayadeva
The 'Gita Govinda', on Krishna and Radha, was written by Jayadeva in the 12th century.

English Language and Comprehension

Q.1 · Answer (D) Contracting
'Bulging' means swelling outwards; its opposite is 'contracting'.

Q.2 · Answer (A) have I
When a sentence opens with the negative 'never', the auxiliary comes before the subject: 'Never have I ever seen ...'

Q.3 · Answer (D) brake
A vehicle is slowed with its 'brake'; 'break' sounds the same but means something else.

Q.4 · Answer (D) Mix
To 'blend' is to 'mix'.

Q.5 · Answer (C) Your father will be waiting for you when you reach home.
An action that will be in progress at a future time takes the future continuous: 'will be waiting'.

Q.6 · Answer (D) tree
To 'bark up the wrong tree' is to look for something, or complain, in the wrong place.

Q.8 · Answer (B) grey
Nothing is being compared, so the positive degree: 'appeared to be grey after the storm'.

Q.9 · Answer (A) principles
Living within environmental limits is one of the central 'principles', the basic ideas, of sustainable development. 'Illuminates' and 'demonstrates' are verbs.

Q.10 · Answer (B) Spectaculor
The correct spelling is 'spectacular'.

Q.11 · Answer (C) broke down
A car that 'breaks down' stops working. To 'break out' is to start suddenly, to 'break in' to enter by force and to 'break up' to come to an end.

Q.12 · Answer (D) bedrock of child survival
'Bedrock' takes 'of': 'the bedrock of child survival'.

Q.13 · Answer (C) heavy
Nothing is being compared, so the positive degree: 'made up of heavy metal'.

Q.14 · Answer (C) to catch out with
To make up for what one has missed is to 'catch up with' it; to 'catch out' is to trap someone in a mistake.

Q.15 · Answer (C) courageous
'Am not ... enough' needs an adjective: 'I am not courageous enough to take this risk.'

Q.16 · Answer (A) Monika said to her mother, "I have passed
Direct speech, the speaker's exact words in quotation marks, is not introduced with 'that'; 'that' belongs to reported speech ('Monika told her mother that she had passed').

Q.18 · Answer (B) gay
Nothing is being compared, so the positive degree: 'the young gay students', 'gay' here meaning cheerful.

Q.19 · Answer (A) Cheerfulness
'Melancholy' is a deep sadness; its opposite is 'cheerfulness'.

Q.20 · Answer (D) Circumfarence
The correct spelling is 'circumference'. 'Kilometers' is the American spelling, not a mistake.

Q.21 · Answer (D) use of
'Make use of' is the fixed phrase: traditional medicines make use of natural products.

Q.22 · Answer (D) employ
The forms of medicine listed are a plural subject in the present tense: they 'employ' natural products.

Q.23 · Answer (B) practised
'Have been' needs a past participle: 'have been practised' (the verb 'practise' in British spelling).

Q.24 · Answer (B) have
The plural 'they' takes 'have' in the present perfect: 'they have blossomed into ... systems of medicine'.

Q.25 · Answer (D) are
The plural 'they' takes 'are': 'they are still a valuable repository of human knowledge'.

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